Task 2: AI-Powered Product Prototype Strategy & Decisions Document

DOST Compass: AI-Powered Stepped Care Triage & Continuity Navigator

Author: Cheekatipalli Nandhini

Role Target: Business Analyst, Founder's Office — YourDOST

Problem Statement: *How might we use AI to help users identify the most relevant next step in their mental wellbeing journey?*

Deliverables Included:

- **1.** Functional Interactive Prototype: Node.js/Express Runnable Web Application (`http://localhost:3000`)
- **2.** High-Resolution Visual Evidence: `prototype_screenshot.png`
- **3.** Strategic Decisions & Product Architecture Document (This document + PDF export)
- **4.** Comprehensive User Journey Walkthrough Guide (`Task2_Prototype_Walkthrough.md`)

---

1. Identifying the Opportunity

The Specific User Problem: Choice Paralysis & Emotional Friction at the Front Door

In a B2B employee assistance platform like YourDOST, users arrive under states of acute cognitive overload, burnout, vulnerability, and pervasive stigma. The current industry paradigm confronts these overwhelmed employees with a "catalog of options": 150+ self-help articles, 40+ DIY audio tracks, multiple clinical assessments (PHQ-9, GAD-7), and a directory of dozens of therapists.

This creates severe **cognitive friction and abandonment**:

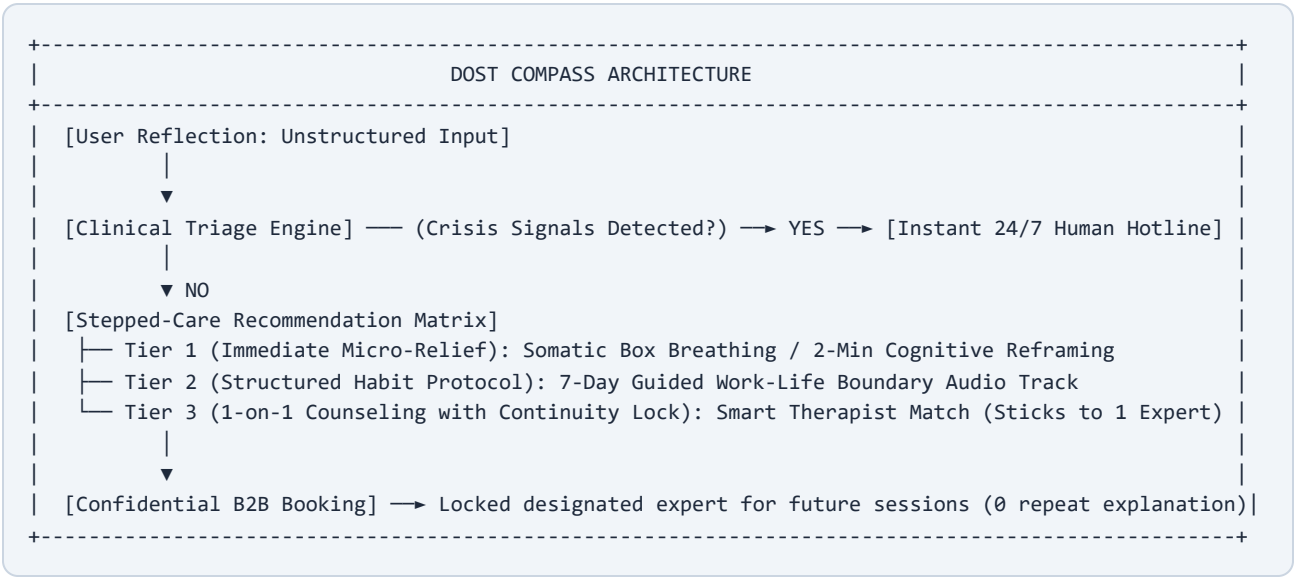
- **1.** **Choice Paralysis:** When someone is emotionally depleted or experiencing anxiety, deciding whether they need a 45-minute therapy session, a 5-minute breathing exercise, or an article induces decision fatigue.
- **2.** **The "Therapy Chasm":** Jumping directly to booking a 1-on-1 counselor feels intimidating, high-commitment, and socially stigmatized. Conversely, generic articles feel trivializing to someone in deep distress.
- **3.** **The Provider Switching Dilemma (YourDOST Core Challenge):** As highlighted in the prompt, YourDOST operates on a model where employees can book any expert, but *the business and clinical goal is for them to stick to one expert*. When users don't know who to choose, they bounce across providers, destroying clinical continuity and therapeutic rapport.

Why AI is the Right Solution

- **Nuanced Natural Language Interpretation:** Emotional distress is expressed messily (*"I feel like a weight is on my chest and sprint deadlines are crushing me"*), which cannot be mapped by rigid dropdown filters. AI parses sentiment, life domain, and severity without forcing the user to take a clinical test.
- **Dynamic Cognitive Load Reduction (Stepped Care):** Instead of presenting a 50-item catalog, AI synthesizes the user's state into **"The Single Best Next Step"** accompanied by a structured 3-tier care stack.
- **Continuous Safety Monitoring & Zero-Latency Triage:** AI provides immediate, deterministic guardrails to detect crisis/self-harm markers and route to 24/7 human clinicians instantly.

---

2. Product Solution & Key Features: "DOST Compass"



Core Features

- **1.** **Safe-Space Empathetic Intake:** Natural conversational interface where employees express feelings freely. No clinical jargon or mandatory questionnaires.
- **2.** **Deterministic Safety Guardrails & SOS Escalation:** Real-time regex and contextual pattern matching for crisis keywords (suicide, self-harm, severe despair). Triggers an immediate, unblockable red alert banner with 24/7 YourDOST SOS, national Tele-MANAS (14416), and Vandrevala Foundation helplines.
- **3.** **Stepped-Care Recommendation Engine:**
- **Level 1 (Immediate Micro-Relief, 2 mins):** Interactive somatic reset embedded directly in the viewport (e.g., visual Box Breathing widget pacing 4–4–4–4 breath cycles).
- **Level 2 (Self-Guided Structured Program):** Micro-learning modules (e.g., *7-Day Workplace Boundary Shield*).
- **Level 3 (1-on-1 Expert Match with "Continuity Lock"):** Algorithmically pairs the user with 1 designated psychologist based on clinical domain, corporate context, and language.
- **4.** **The "Continuity Lock" Guarantee:** Explicitly solves YourDOST's operational mandate. Reassures the user: *"Meet Dr. Ananya Sen. She is locked as your dedicated care partner for all future check-ins so you never have to re-explain your backstory."*
- **5.** **Clinical Ops & HR Insights Drawer:** Gives internal clinical response teams real-time visibility into triage volumes and severity streams, while providing corporate HR with anonymized, aggregated burnout indices.

---

3. Critical Product Decisions, Assumptions & Trade-Offs

Key Assumptions

- **1.** **B2B Trust Ceiling:** Employees will only share vulnerable truths if they are 100% convinced their employer/HR has zero access to individual logs.
- **2.** **Willingness for Low-Stakes Engagement:** A burnt-out employee who declines a 45-minute counseling session will readily engage with a 2-minute somatic breathing exercise, which subsequently lowers the psychological threshold for booking therapy.
- **3.** **Continuity Driver:** Sticking to one counselor produces higher Net Promoter Score (NPS) and clinical symptom improvement (PHQ-4 / GAD-7 delta) than open-market therapist shopping.

Strategic Trade-Offs Made

- **Stepped Recommendation vs. Full Catalog Access:**

- *Decision:* Limit initial view to **one primary recommendation** and a 3-tier care stack, rather than displaying the complete library of 100+ resources.
- *Trade-Off:* Sacrifice broad content discovery in exchange for slashing cognitive load and decision paralysis.
- **Deterministic Crisis Gating vs. LLM Freedom:**
- *Decision:* In acute distress scenarios, the AI is **hard-coded** to halt open conversational therapy and redirect immediately to human emergency hotlines.
- *Trade-Off:* The AI never attempts to "counsel" a suicidal or actively self-harming individual. Safety and liability strictly supersede engagement.
- **Rule-Based Hybrid Classifier vs. Pure Black-Box LLM:**
- *Decision:* Utilize structured clinical domain heuristics coupled with sentiment embeddings rather than an unconstrained generative chatbot.
- *Trade-Off:* Guarantees predictable, auditable clinical outcomes and eliminates hallucinations.

---

4. Handling AI Errors, Privacy & Crisis Escalation

Risk Area	Failure Mode	Mitigation & System Protocol
Crisis & Red Flags	User expresses self-harm or acute suicidality.	**Zero-Tolerance Human Escalation:** Instant UI takeover. 1-click connection to YourDOST 24/7 on-call supervisor; logs clinical alert; suppresses algorithmic generation.
AI Misclassification	Mild stress misdiagnosed as severe, or vice-versa.	**Stepped Fallback:** All tiers remain visible on screen. If a user feels counseling is too intensive, they can effortlessly pivot to the Level 1 breathing tool or Level 2 DIY article with zero friction.
B2B Privacy & Stigma	Employee fears HR retaliation or surveillance.	**Zero Corporate Data Sharing:** Data architecture isolates corporate employer billing IDs from clinical conversation logs. HR dashboards display only k-anonymized aggregate indices ($N \ge 50$).
Therapist Misalignment	User does not feel rapport with matched counselor.	**Frictionless Re-Matching:** While the system defaults to the "Continuity Lock", the user is given a gentle 1-click option: *"Request a different specialist"* without awkwardness.

---

5. Success Metrics & Evaluation Framework

To evaluate whether DOST Compass successfully solves the user problem and aligns with YourDOST's business objectives, we define a balanced scorecard:

```
+-----------------------------+-----------------------------------------------------------------+
| Metric Category             | Specific KPI & Target Benchmark                                 |
+-----------------------------+-----------------------------------------------------------------+
| 1. Funnel Activation Rate   | % of arriving users who complete a "Next Step" within 3 mins    |
|                             | Benchmark: Increase from 22% (baseline) to > 55%                |
+-----------------------------+-----------------------------------------------------------------+
| 2. Expert Stickiness Rate   | % of counseling clients who book Session 2+ with SAME expert    |
| (YourDOST Core Goal)        | Target: > 75% continuity retention (vs. industry 38%)           |
+-----------------------------+-----------------------------------------------------------------+
| 3. Clinical Symptom Delta   | Pre-post distress score change (PHQ-4 / Burnout Index)          |
|                             | Target: Statistically significant drop after 3 touchpoints      |
+-----------------------------+-----------------------------------------------------------------+
| 4. Safety Audit Accuracy    | Zero missed crisis escalations (100% sensitivity for SOS)       |
```

```
|                          | False positive rate < 3%                                                                  |
+——————————————————————————+———————————————————————————————————————————————————————————————————————————————+
```